



IMATERO 100 (Imatinib Tablets 100mg)
IMATERO 400 (Imatinib Tablets 400mg)

ACTIVE INGREDIENT

IMATERO 100
Each film-coated tablet contains 100mg imatinib.
IMATERO 400
Each film-coated tablet contains 400mg imatinib.

DOSAGE FORM

Oral film coated Tablets

PRODUCT DESCRIPTIONS

IMATERO 100
White to off white coloured, round, bevel edged scored film coated tablets, debossed with H on one side and 19 on the other side, 1 and 9 separated by a score line.
IMATERO 400
White to off white coloured, capsule shaped, bevel edged scored film coated tablets, debossed with H on one side and 20 on the other side, 2 and 0 separated by a score line.

PHARMACODYNAMICS

Pharmacotherapeutic group: protein-tyrosine kinase inhibitor, ATC code: L01EA01
Mechanism of action
Imatinib is a small molecule protein-tyrosine kinase inhibitor that potently inhibits the activity of the BCR-ABL tyrosine kinase (TK), as well as several receptor TKs: KIT, the receptor for stem cell factor (SCF) coded for by the c-KIT proto-oncogene, the discoidin domain receptors (DDR1) and DDR2), the colony stimulating factor receptor (CSF-1R) and the platelet-derived growth factor receptors alpha and beta (PDGFR-alpha and PDGFR-beta). Imatinib can also inhibit cellular events mediated by activation of these receptor kinases.
Pharmacodynamic effects
Imatinib is a protein-tyrosine kinase inhibitor, which potently inhibits the breakpoint cluster region-Abelson (BCR-ABL) tyrosine kinase at the in vitro, cellular, in vivo levels. The compound selectively inhibits proliferation and induces apoptosis in BCR-ABL positive cell lines as well as fresh leukemic cells from Philadelphia chromosome positive CML and acute lymphoblastic leukemia (ALL) patients. In colony transformation assays using ex vivo peripheral blood and bone marrow samples, imatinib shows selective inhibition of BCR-ABL positive colonies from CML patients.
In vivo the compound shows anti-tumour activity as a single agent in animal models using BCR-ABL positive tumour cells. Imatinib is also an inhibitor of the receptor tyrosine kinases for platelet-derived growth factor (PDGF) and stem cell factor (SCF), KIT, and inhibits PDGF- and SCF-mediated cellular events. In vitro, imatinib inhibits proliferation and induces apoptosis in gastrointestinal stromal tumour (GIST) cells, which express an activating KIT mutation. Constitutive activation of the PDGFR or the ABL protein tyrosine kinases as a consequence of fusion to diverse partner proteins or constitutive production of PDGF have been implicated in the pathogenesis of MDS/MPD, HES/CEL and DFSP. In addition, constitutive activation of KIT or the PDGFR has been implicated in the pathogenesis of ASM. Imatinib inhibits signalling and proliferation of cells driven by dysregulated PDGFR, KIT and ABL kinase activity.

PHARMACOKINETIC PROPERTIES

Absorption
Mean absolute bioavailability for imatinib is 98%. There was high between-patient variability in plasma imatinib AUC levels after an oral dose. When given with a high-fat meal, the rate of absorption of imatinib was minimally reduced (11% decrease in Cmax and prolongation of tmax by 1.5 h), with a small reduction in AUC (7.4%) compared to fasting conditions.
Distribution
At clinically relevant concentrations of imatinib, binding to plasma proteins was approximately 95% on the basis of *in vitro* experiments, mostly to albumin and alpha-acid-glycoprotein, with little binding to lipoprotein.
Biotransformation
The main circulating metabolite in humans is the N-demethylated piperazine derivative (CGP71588), which shows similar *in vitro* potency as the parent compound. The plasma AUC for this metabolite was found to be only 16% of the AUC for imatinib. The plasma protein binding of the N-demethylated metabolite is similar to that of the parent compound.
Elimination
Based on the recovery of compound(s) after an oral ¹⁴C-labelled dose of imatinib, approximately 81% of the dose was recovered within 7 days in faeces (68% of dose) and urine (13% of dose). Unchanged imatinib accounted for 25% of the dose (5% urine, 20% faeces), the remainder being metabolites.
The mean apparent elimination half-time is 13.5 hours. The half-life of all ¹⁴C-labelled components in plasma is from 41-72 hours.
Plasma pharmacokinetics
The t½ is approximately 18 h, suggesting that once-daily dosing is appropriate. The increase in mean AUC with increasing dose is linear and dose proportional in the range of 25- 1,000 mg/imatinib after oral administration. There was no change in the kinetics of imatinib on repeated dosing, and accumulation was 1.5-2.5-fold at steady state when dosed once daily.
Special populations
Based on population pharmacokinetic analysis, there was a small effect of age on the volume of distribution (12% increase in patients >65 years old). This change is not thought to be clinically significant. The effect of body weight on the clearance of imatinib is such that for a patient weighing 50 kg the mean clearance is expected to be 8.5 L/h, while for a patient weighing 100 kg the clearance will rise to 11.8 L/h. These changes are not considered sufficient to warrant dose adjustment based on kg bodyweight. There is no effect of gender on the kinetics of imatinib.
Further population PK analysis in the phase III study in newly diagnosed CML patients showed that the effect of covariates and co-medications on both clearance and volume of distribution appears to be small and is not sufficiently pronounced to warrant dose adjustment.
Pediatric patients (below 18 years)
As in adult patients, imatinib was rapidly absorbed after oral administration in pediatric patients. Dosing in children at 260 and 340 mg/m² achieved the same exposure, respectively, as doses of 400 mg and 600 mg in adult patients. The comparison of AUC(0-24) on Day 8 and Day 1 at 340 mg/m² dose level revealed a 1.7 fold drug accumulation after repeated once daily dosing. Clearance of imatinib increases with increasing body surface area (BSA). After correcting for the BSA effect, other demographics such as age, body weight and body mass index did not have clinically significant effects on the exposure of imatinib. The analysis confirmed that exposure of imatinib in pediatric patients receiving 260 mg/m² once daily (not exceeding 400 mg once daily) or 340 mg/m² once daily (not exceeding 600 mg once daily) were similar to those in adult patients who received imatinib 400 mg or 600 mg once daily.

Organ function impairment
Imatinib and its metabolites are not excreted via the kidney to a significant extent. Patients with mild and moderate impairment of renal function appear to have a higher plasma exposure than patients with normal renal function. The increase is approximately 1.5- to 2-fold, corresponding to a 1.5-fold elevation of plasma AGP, to which imatinib binds strongly. The free drug clearance of imatinib is probably similar between patients with renal impairment and those with normal renal function, since renal excretion represents only a minor elimination pathway for imatinib.
Although the results of pharmacokinetic analysis showed that there is considerable inter-subject variation, the mean exposure to imatinib did not increase in patients with varying degrees of liver dysfunction as compared to patients with normal liver function.

INDICATION

IMATERO is indicated for the treatment of:

- adult and pediatric patients with newly diagnosed chronic myeloid leukemia (CML) as well as for the treatment of adult and pediatric patients with CML in blast crisis, accelerated phase, or in chronic phase after failure of interferon-alpha therapy.
- adult patients with unresectable and/or metastatic malignant gastrointestinal stromal tumours (GIST).
- adjuvant treatment of adult patients following resection of GIST. Patients who have a low or very low risk of recurrence should not receive adjuvant treatment.
- adult and pediatric patients with newly diagnosed Philadelphia chromosome positive acute lymphoblastic leukemia (Ph+ ALL) integrated with chemotherapy.
- adult patients with relapsed or refractory Ph+ ALL as monotherapy.
- adult patients with myelodysplastic/myeloproliferative diseases (MDS/MPD) associated with platelet-derived growth factor receptor (PDGFR) gene re-arrangements.
- adult patients with hyper eosinophilic syndrome (HES) and/or chronic eosinophilic leukemia (CEL) with FIP1L1-PDGFRα rearrangement.
- adult patients with unresectable, recurrent and/or metastatic dermatofibrosarcoma protuberans (DFSP).
- adult patients with aggressive systemic mastocytosis (ASM) without the D816V c-Kit mutation or with c-Kit mutational status unknown.

The effectiveness of IMATERO is based on overall haematological and cytogenetic response rates and progression-free survival in CML, on haematological and cytogenetic response rates in Ph+ ALL, MDS/MPD, on haematological response rates in HES/CEL and ASM and on objective response rates in GIST and DFSP, and on recurrence-free survival in adjuvant GIST (see section PHARMACODYNAMICS). The experience with IMATERO in patients with MDS/MPD associated with PDGFR gene re-arrangements is very limited. Except in newly diagnosed chronic phase CML, there are no controlled trials demonstrating a clinical benefit or increased survival in diseases.

RECOMMENDED DOSE

Therapy should be initiated by a physician experienced in the treatment of patients with hematological malignancies and malignant sarcomas, as appropriate.
The prescribed dose should be administered orally with a meal and a large glass of water to minimize the risk of gastrointestinal disturbances. Doses of 400 mg or 600 mg should be administered once daily, whereas a daily dose of 800 mg should be administered as 400 mg twice a day, in the morning and in the evening.
For patients unable to swallow the film-coated tablets, the tablets may be dispersed in a glass of water or apple juice. The required number of tablets should be placed in the appropriate volume of beverage (approximately 50 mL for a 100 mg tablet, and 200 mL for a 400 mg tablet) and stirred with a spoon. The suspension should be administered immediately after complete disintegration of the tablet(s).
Treatment should be continued as long as the patient continues to benefit.
Monitoring of response to IMATERO therapy in Ph+ CML patients should be performed routinely and when therapy is modified, to identify suboptimal response, loss of response to therapy, poor patient compliance, or possible drug-drug interaction. Results of monitoring should guide appropriate CML management.

General target population:

Dosage in CML

The recommended dosage of IMATERO is 400 mg/day for adult patients in chronic phase CML and 600 mg/day for patients in accelerated phase or blast crisis.
Dose increase from 400 mg to 600 mg or 800 mg in patients with chronic phase disease, or from 600 mg to a maximum of 800 mg daily in patients in accelerated phase or blast crisis may be considered in the absence of severe adverse drug reaction and severe non-leukemia-related neutropenia or thrombocytopenia in the following circumstances: disease progression (at any time); failure to achieve a satisfactory hematological response after at least 3 months of treatment; failure to achieve a cytogenetic response after 12 months of treatment; or loss of a previously achieved hematological and/or cytogenetic response.
See section on special populations for pediatric patients.

Dosage in Ph+ ALL

The recommended dose of IMATERO is 600 mg/day for adult patients with Ph+ ALL. See section on special populations for pediatric patients.

Dosage in MDS/MPD

The recommended dose of IMATERO is 400 mg/day for adult patients with MDS/MPD.

Dosage in ASM

The recommended dose of IMATERO is 400 mg/day for adult patients with ASM without the D816V c-KIT mutation or mutational status unknown or not responding satisfactorily to other therapies.
For patients with ASM associated with eosinophilia, a clonal hematological disease related to the fusion kinase FIP1L1-PDGFR-alpha, a starting dose of 100 mg/day is recommended. A dose increase from 100 mg to 400 mg for these patients may be considered in the absence of adverse drug reactions if assessments demonstrate an insufficient response to therapy.

Dosage in HES/CEL

The recommended dose of IMATERO is 400 mg/day for adult patients with HES/CEL.
For HES/CEL patients with demonstrated FIP1L1-PDGFR-alpha fusion kinase, a starting dose of 100 mg/day is recommended. A dose increase from 100 mg to 400 mg for these patients may be considered in the absence of adverse drug reactions if assessments demonstrate an insufficient response to therapy.

Dosage in GIST

The recommended dose of IMATERO is 400 mg/day for adult patients with unresectable and/or metastatic, malignant GIST.

A dose increase from 400 mg to 600 mg or 800 mg for patients may be considered in the absence of adverse drug reactions if assessments demonstrate an insufficient response to therapy.
The recommended dose of IMATERO is 400 mg/day for the adjuvant treatment of adult patients following complete gross resection of GIST. The optimal treatment duration with IMATERO is not known.

Dosage in DFSP

The recommended dose of IMATERO is 800 mg/day for adult patients with DFSP.

Dose adjustments for adverse drug reactions:

Non-hematological adverse drug reactions

If a severe non-hematological adverse drug reaction develops with IMATERO use, treatment must be withheld until the event has resolved. Thereafter, treatment can be resumed as appropriate depending on the initial severity of the event.
If elevations in bilirubin >3 x institutional upper limit of normal (IULN) or in liver transaminases >5 x IULN occur, IMATERO should be withheld until bilirubin levels have returned to a <1.5 x IULN and transaminase levels to <2.5 x IULN. Treatment with IMATERO may then be continued at a reduced daily dose. In adults the dose should be reduced from 400 to 300 mg, or from 600 to 400 mg, or from 800 mg to 600 mg, and in pediatric patients from 340 to 260 mg/m²/day.

Hematological adverse drug reactions

Dose reduction or treatment interruption for severe neutropenia and thrombocytopenia are recommended as indicated in the table below.

Table 1 Dose adjustments for neutropenia and thrombocytopenia

ASM associated with eosinophilia and HES/CEL withFIP1L1-PDGFR-alpha fusion kinase (starting dose 100 mg)	ANC < 1.0 x 10 ⁹ /L and/or platelets < 50 x 10 ⁹ /L	1. Stop IMATERO until ANC ≥ 1.5 x 10 ⁹ /L and platelets ≥ 75 x 10 ⁹ /L 2. Resume treatment with IMATERO at previous dose(i.e., before severe adverse drug reaction)
Chronic phase CML, MDS/MPD, ASM, HES/CEL andGIST (starting dose 400 mg)	ANC < 1.0 x10 ⁹ /L and/or platelets < 50 x10 ⁹ /L	1. Stop IMATERO until ANC ≥ 1.5 x10 ⁹ /L and platelets ≥ 75 x10 ⁹ /L. 2. Resume treatment with IMATERO at previous dose(i.e., before severe adverse drug reaction). 3. In the event of recurrence of ANC < 1.0 x10 ⁹ /L and/or platelets < 50 x10 ⁹ /L, repeat step 1 and resume IMATERO at reduced dose of 300 mg.
Pediatric chronic phase CML(at dose 340 mg/m ²)	ANC < 1.0 x10 ⁹ /L and/or platelets < 50 x10 ⁹ /L	1. Stop IMATERO until ANC ≥ 1.5 x 10 ⁹ /L and platelets ≥ 75 x 10 ⁹ /L. 2. Resume treatment with IMATERO at previous dose (i.e., before severe adverse drug reaction) 3. In the event of recurrence of ANC < 1.0 x 10 ⁹ /L and/or platelets < 50 x 10 ⁹ /L, repeat step 1 and resume IMATERO at reduced dose of 260 mg/m ² .
Accelerated phase CML and blast crisis and Ph+ ALL(starting dose 600 mg ^a)	^a ANC < 0.5 x 10 ⁹ /L and/or platelets < 10 x10 ⁹ /L	1. Check whether cytopenia is related to leukemia (marrow aspirate or biopsy). 2. If cytopenia is unrelated to leukemia, reduce dose ofIMATERO to 400 mg ^b . 3. If cytopenia persists for 2 weeks, reduce further to 300 mg ^d . 4. If cytopenia persists for 4 weeks and is still unrelated to leukemia, stop IMATERO until ANC ≥ 1 x 10 ⁹ /L and platelets ≥ 20 x 10 ⁹ /L, then resume treatment at 300 mg ^d .
DFSP (starting dose 800 mg)	ANC < 1.0 x10 ⁹ /L and/or platelets < 50 x10 ⁹ /L	1. Stop IMATERO until ANC ≥ 1.5 x 10 ⁹ /L and platelets ≥ 75 x 10 ⁹ /L. 2. Resume treatment with IMATERO at 600 mg 3. In the event of recurrence of ANC < 1.0 x 10 ⁹ /L and/or platelets < 50 x10 ⁹ /L, repeat step 1 and resume IMATERO at reduced dose of 400 mg.
ANC = absolute neutrophil count ^a occurring after at least 1 month of treatment ^b or 260 mg/m ² in pediatric patients ^c or 340 mg/m ² in pediatric patients ^d or 200 mg/m ² in pediatric patients		

Special populations:

Renal insufficiency

Imatinib and its metabolites are not significantly excreted via the kidney. Patients with renal dysfunction or on dialysis could be given the minimum recommended dose of 400 mg daily as starting dose. However, in these patients caution is recommended. The dose can be reduced if not tolerated. If tolerated, the dose can be increased for lack of efficacy.

Hepatic impairment

Imatinib is mainly metabolized by the liver. Patients with mild, moderate or severe liver impairment should be given the minimum recommended dose of 400 mg daily. The dose can be reduced if not tolerated.

Pediatric patients (below 18 years)

There is no experience with the use of imatinib in pediatric patients with CML below 2 years of age and with Ph+ALL below 1 year of age. There is very limited to no experience with the use of imatinib in pediatric patients in other indications. Dosing in pediatric patients should be on the basis of body surface area (mg/m²). The dose of 340 mg/m² daily is recommended for children with chronic phase and advanced phase CML and Ph+ALL (not to exceed the total dose of 600 mg daily). Treatment can be given as a once daily dose in CML and Ph+ALL.
In CML, alternatively the daily dose may be split into two administrations – one in the morning and one in the evening.

Geriatric patients (65 years or above)

No significant age-related pharmacokinetic differences have been observed in adult patients. No specific dose recommendation is necessary in the elderly.

MODE OF ADMINISTRATION

Oral route of administration

CONTRAINDICATIONS

Use in patients with a hypersensitivity to the active substance or to any of the excipients.

WARNINGS AND PRECAUTIONS

When imatinib tablets is co-administered with other medicinal products, there is a potential for drug interactions. Caution should be used when taking imatinib tablets with protease inhibitors, azole antifungals, certain macrolides, CYP3A4 substrates with a narrow therapeutic window (e.g., cyclosporine or pimozide) or CYP2C9 substrates with a narrow therapeutic window (e.g., warfarin and other coumarin derivatives).

Hypothyroidism

Clinical cases of hypothyroidism have been reported in thyroidectomy patients undergoing levothyroxine replacement during treatment with imatinib tablets. Thyroid-stimulating hormone (TSH) levels should be closely monitored in such patients.

Hepatotoxicity

In patients with hepatic dysfunction (mild, moderate or severe), peripheral blood counts and liver enzymes should be carefully monitored.
When imatinib tablets is combined with high dose chemotherapy regimens, transient liver toxicity in the form of transaminase elevation and hyperbilirubinemia has been observed. Additionally, there have been uncommon reports of acute liver failure. Monitoring of hepatic function is recommended in circumstances where imatinib tablets is combined with chemotherapy regimens also known to be associated with hepatic dysfunction.

Fluid retention

Occurrences of severe fluid retention (pleural effusion, oedema, pulmonary oedema, ascites, superficial oedema) have been reported in approximately 2.5% of newly diagnosed CML patients taking imatinib tablets. Therefore, it is highly recommended that patients be weighed regularly. An unexpected rapid weight gain should be carefully investigated and if necessary appropriate supportive care and therapeutic measures should be undertaken. Caution should be exercised in patients with cardiac dysfunction.

Patients with cardiac disease or renal failure

Patients with cardiac disease, risk factors for cardiac failure or history of renal failure should be monitored carefully and any patient with signs or symptoms consistent with cardiac or renal failure should be evaluated and treated.
In patients with hyper eosinophilic syndrome (HES) with occult infiltration of HES cells within the myocardium, isolated cases of cardiogenic shock/left ventricular dysfunction have been associated with HES cell degranulation upon the initiation of imatinib tablets therapy. The condition was reported to be reversible with the administration of systemic steroids, circulatory support measures and temporarily withholding imatinib tablets. Myelodysplastic (MDS)/myeloproliferative diseases (MPD) and systemic mastocytosis might be associated with high eosinophil levels. Performance of an echocardiogram and determination of serum troponin should therefore be considered in patients with HES/CEL, and in patients with MDS/MPD or ASM associated with high eosinophil levels. If either is abnormal, the prophylactic use of systemic steroids (1 to 2 mg/kg) for one to two weeks concomitantly with imatinib tablets should be considered at the initiation of therapy.

Gastrointestinal haemorrhage

Patients should be monitored for gastrointestinal symptoms at the start of and during therapy with imatinib tablets. When needed, imatinib tablets discontinuation may be considered.

Tumor lysis syndrome

Cases of tumor lysis syndrome (TLS) have been reported in patients treated with imatinib tablets. Due to possible occurrence of TLS, correction of clinically significant dehydration and treatment of high uric acid levels are recommended prior to initiation of imatinib tablets.

Hepatitis B reactivation

Reactivation of hepatitis B can occur in patients who are chronic carriers of this virus after receiving a BCR-ABL tyrosine kinase inhibitor (TKi), such as imatinib. Some cases involving drugs of the BCR-ABL TKi class resulted in acute hepatic failure or fulminant hepatitis leading to liver transplantation or a fatal outcome.
Patients should be tested for hepatitis B infection before initiating treatment with imatinib. Patients currently on imatinib should have baseline testing for hepatitis B infection in order to identify chronic carriers of the virus. Experts in liver disease and in the treatment of hepatitis B should be consulted before treatment is initiated in patients with positive hepatitis B serology (including those with active disease) and for patients who test positive for hepatitis B infection during treatment. Carriers of hepatitis B virus who require treatment with imatinib should be closely monitored for signs and symptoms of active hepatitis B infection throughout therapy and for several months following termination of therapy.

Thrombotic microangiopathy

BCR-ABL tyrosine kinase inhibitors (TKIs) have been associated with thrombotic microangiopathy (TMA), including individual case reports for Imatinib. If laboratory or clinical findings associated with TMA occur in a patient receiving Imatinib, treatment should be discontinued and thorough evaluation for TMA, including ADAMTS13 activity and anti-ADAMTS13-antibody determination, should be completed. If anti-ADAMTS13-antibody is elevated in conjunction with low ADAMTS13 activity, treatment with Imatinib should not be resumed.

Laboratory tests

Haematology

Complete blood counts must be performed regularly during therapy with imatinib tablets. Treatment of CML patients with imatinib tablets has been associated with neutropenia or thrombocytopenia. However, the occurrence of these cytopenias is dependent on the stage of the disease being treated and they were more frequent in patients with accelerated phase CML or blast crisis as compared to patients with chronic phase CML. Treatment with imatinib may be interrupted or the dose be reduced.

Liver function

Liver function (transaminases, bilirubin, alkaline phosphatase) should be monitored regularly in patients receiving imatinib tablets.

Size : 280 x 550 mm

Pharma Code: 9506, Folding Size: 35 x 55 mm

Spec: Printed on 40 GSM Bible paper, front & back side printing.

Note: Pharma code position and Orientation are tentative, will be changed according to printers

requirement to suit pharma code position at centre after folding.

Colour (01): Black

These laboratory abnormalities should be managed with interruption and/or dose reduction of the treatment with imatinib tablets.

Renal function

Imatinib and its metabolites are not excreted via the kidney to a significant extent. Creatinine clearance (CrCL) is known to decrease with age, and age did not significantly affect imatinib kinetics. In patients with impaired renal function, imatinib plasma exposure seems to be higher than that in patients with normal renal function, probably due to an elevated plasma level of alpha-acid glyco-protein (AGP), an imatinib-binding protein, in these patients. There is no correlation between imatinib exposure and the degree of renal impairment, as classified by the measurement of creatinine clearance (CrCL), between patients with mild (CrCL: 40 to 59 mL/min) and severe (CrCL: <20 mL/min) renal impairment. However, the starting dose of imatinib can be reduced if not tolerated. Long-term treatment with imatinib tablets may be associated with a clinically significant decline in renal function. Renal function should, therefore, be evaluated prior to the start of imatinib tablets therapy and closely monitored during therapy, with particular attention to those patients exhibiting risk factors for renal dysfunction. If renal dysfunction is observed, appropriate management and treatment should be initiated in accordance with standard treatment guidelines.

Pediatric patients (below 18 years)

There have been case reports of growth retardation occurring in children and pre-adolescents receiving imatinib tablets. The long-term effects of prolonged treatment with imatinib tablets on growth in children are unknown. Therefore, close monitoring of growth in children under imatinib treatment is recommended.

INTERACTION WITH OTHER MEDICINAL PRODUCTS AND OTHER FORMS OF INTERACTION

Observed interactions resulting in a concomitant use not recommended

Durgs that may decrease imatinib plasma concentrations:

Substances that are inducers of CYP3A4 activity (e.g., dexamethasone, phenytoin, carbamazepine, rifampicin, phenobarbital or hypericum perforatum, also known as St. John’s wort) may significantly reduce exposure to imatinib. In patients where rifampicin or other CYP3A4 inducers are indicated, alternative therapeutic agents with less enzyme induction potential should be considered.

Other interactions that may affect exposure to IMATERO or other drugs

Drugs that may increase imatinib plasma concentrations:

Substances that inhibit the cytochrome P450 isoenzyme CYP3A4 activity (e.g., ketoconazole, itraconazole, erythromycin, clarithromycin) could decrease metabolism and increase imatinib concentrations. There was a significant increase in exposure to imatinib (the mean Cmax and AUC of imatinib rose by 26% and 40%, respectively) in healthy subjects when it was co-administered with a single dose of ketoconazole (a CYP3A4 inhibitor). Caution should be taken when administering imatinib with inhibitors of the CYP3A4 family.

Drugs that may have their plasma concentration altered by imatinib:

Imatinib increases the mean Cmax and AUC of simvastatin (CYP3A4 substrate) 2- and 3.5-fold, respectively, indicating an inhibition of the CYP3A4 by imatinib. Therefore, caution is recommended when administering imatinib with CYP3A4 substrates with a narrow therapeutic window (e.g., cyclosporine, pimozide, tacrolimus, sirolimus, ergotamine, diergotamine, fentanyl, alfentanil, terfenadine, bortezomib, docetaxel and quindidine). Imatinib may increase plasma concentration of other CYP3A4 metabolised medicinal products (e.g., triazolo-benzodiazepines, dihydropyridine calcium channel blockers, certain HMG-CoA reductase inhibitors, i.e., statins, etc.).

Imatinib also inhibits CYP2C9 and CYP2C19 activity in vitro. PT prolongation was observed following co-administration with warfarin. When giving coumarins, short-term PT monitoring is therefore necessary at the start and end of imatinib therapy and when altering the dosage. Alternatively, the use of low-molecular weight heparin should be considered.

In vitro, imatinib inhibits the cytochrome P450 isoenzyme CYP2D6 activity at concentrations similar to those that affect CYP3A4 activity. Imatinib at 400 mg twice daily had a weak inhibitory effect on CYP2D6-mediated metoprolol metabolism, with metoprolol Cmax and AUC being increased by approximately 23%. Co-administration of imatinib with CYP2D6 substrates, such as metoprolol, does not seem to be a risk factor for drug-drug interactions and dose adjustment may not be necessary.

In vitro, imatinib tablets inhibits the acetaminophen O-glucuronidate pathway (Ki 58.5 microM).

Co-administration of imatinib (400 mg/day for eight days) with acetaminophen/paracetamol (1000 mg single dose on day eight) in patients with CML did not result in any changes in the pharmacokinetics of acetaminophen/paracetamol.

Imatinib tablets pharmacokinetics was not altered in the presence of single-dose acetaminophen/ paracetamol.

There is no PK or safety data on the concomitant use of imatinib at doses >400 mg/day or the chronic use of concomitant acetaminophen/paracetamol and imatinib.

PREGNANCY AND LACTATION

Pregnancy

IMATERO should be used during pregnancy only if the expected benefit outweighs the potential risk to the fetus. If it is used during pregnancy, the patient must be informed of the potential risk to the fetus.

Lactation

Both imatinib and its active metabolite can be transferred into human milk. The effects of low-dose exposure of the infant to imatinib are unknown, because of the potential for serious adverse drug reactions in the breastfed child, breastfeeding is not recommended during treatment and for at least 15 days after stopping treatment with IMATERO.

Females and males of reproductive potential

Females

Females of reproductive potential should be advised to use effective contraception (methods that result in less than 1 % pregnancy rates) when using IMATERO during treatment and for at least 15 days after stopping treatment with IMATERO.

Infertility

Human studies on male patients receiving imatinib and its effect on male fertility and spermatogenesis have not been performed. Male patients concerned about their fertility on IMATERO treatment should consult with their physician.

SIDE EFFECTS

Infections and infestation	
Uncommon:	Herpes zoster, herpes simplex, nasopharyngitis, pneumonia ¹ , sinusitis, cellulitis, upper respiratory tract infection, influenza, urinary tract infection, gastroenteritis, sepsis
Rare:	Fungal infection
Blood and lymphatic system disorders	
Very common:	Neutropenia, thrombocytopenia, anaemia
Common:	Pancytopenia, febrile neutropenia
Uncommon:	Thrombocythaemia, lymphopenia, bone marrow depression, eosinophilia, lymphadenopathy
Rare	Haemolytic anaemia
Metabolism and nutrition disorders	
Common:	Anorexia
Uncommon:	Hypokalaemia, increased appetite, hypophosphataemia, decreased appetite, dehydration, gout, hyperuricaemia, hypercalcaemia, hyperglycaemia, hyponatraemia
Rare:	Hyperkalaemia, hypomagnesaemia
Psychiatric disorders	
Common:	Insomnia
Uncommon:	Depression, libido decreased, anxiety
Rare:	Confusional state
Nervous system disorders	
Very common:	Headache ²
Common:	Dizziness, paraesthesia, taste disturbance, hypoaesthesia
Uncommon:	Migraine, somnolence, syncope, peripheral neuropathy, memory impairment, sciatica, restless leg syndrome, tremor, cerebral haemorrhage
Rare:	Increased intracranial pressure, convulsions, optic neuritis
Eye disorders	
Common:	Eyelid oedema, lacrimation increased, conjunctival haemorrhage, conjunctivitis, dry eye, blurred vision
Uncommon:	Eye irritation, eye pain, orbital oedema, scleral haemorrhage, retinal haemorrhage, blepharitis, macular oedema
Rare:	Cataract, glaucoma, papilloedema
Ear and labyrinth disorders	
Uncommon:	Vertigo, tinnitus, hearing loss
Cardiac disorders	
Uncommon:	Palpitations, tachycardia, cardiac failure, congestive ³ , pulmonary oedema
Rare:	Arrhythmia, atrial fibrillation, cardiac arrest, myocardial infarction, angina pectoris, pericardial effusion
Vascular disorders ⁴	
Common:	Flushing, haemorrhage
Uncommon:	Hypertension, haematoma, subdural haematoma, peripheral coldness, hypotension, Raynaud’s phenomenon
Respiratory, thoracic and mediastinal disorders	
Common:	Dyspnoea, epistaxis, cough
Uncommon:	Pleural effusion ⁵ , pharyngolaryngeal pain, pharyngitis
Rare:	Pleuritic pain, pulmonary fibrosis, pulmonary hypertension, pulmonary haemorrhage
Gastrointestinal disorders	
Very common:	Nausea, diarrhoea, vomiting, dyspepsia, abdominal pain ⁶
Common:	Flatulence, abdominal distension, gastro-oesophageal reflux, constipation, dry mouth, gastritis
Uncommon:	Stomatitis, mouth ulceration, gastrointestinal haemorrhage ⁷ , eructation, melena, oesophagitis, ascites, gastric ulcer, haematemesis, cheilitis, dysphagia, pancreatitis
Rare:	Colitis, ileus, inflammatory bowel disease
Hepatobiliary disorders	
Common:	Increased hepatic enzymes
Uncommon:	Hyperbilirubinaemia, hepatitis, jaundice
Rare:	Hepatic failure ⁸ , hepatic necrosis ⁹
Skin and subcutaneous tissue disorders	
Very common:	Periorbital oedema, dermatitis/eczema/rash
Common:	Pruritus, face oedema, dry skin, erythema, alopecia, night sweats, photosensitivity reaction
Uncommon:	Rash pustular, contusion, sweating increased, urticaria, ecchymosis, increased tendency to bruise, hypotrichosis, skin hypopigmentation, dermatitis exfoliative, onychoclasis, folliculitis, petechiae, psoriasis, purpura, skin hyperpigmentation, bullous eruptions

Rare:	Acute febrile neutrophilic dermatosis (Sweet’s syndrome), nail discolouration, angioneurotic oedema, rash vesicular, erythema multiforme, leucocytoclastic vasculitis, Stevens-Johnson syndrome, acute generalised exanthematous pustulosis (AGEP)
Musculoskeletal and connective tissue disorders	
Very common:	Muscle spasm and cramps, Musculoskeletal pain including myalgia, arthralgia, bone pain ⁸
Common:	Joint swelling
Uncommon:	Joint and muscle stiffness
Rare:	Muscle weakness, arthritis
Renal and urinary disorders	
Uncommon:	Renal pain, haematuria, renal failure acute, urinary frequency increased
Reproductive system and breast disorders	
Uncommon:	Gynaecomastia, erectile dysfunction, menorrhagia, menstruation irregular, sexual dysfunction, nipple pain, breast enlargement, scrotal oedema
General disorders and administration site conditions	
Very common:	Fluid retention and oedema, fatigue
Common:	Weakness, pyrexia, anasarca, chills, rigors
Uncommon:	Chest pain, malaise
Investigations	
Very common:	Weight increased
Common:	Weight decreased
Uncommon:	Blood creatinine increased, blood creatine phosphokinase increased, blood lactate dehydrogenase increased, blood alkaline phosphatase increased
Rare:	Blood amylase increased
¹ Pneumonia was reported most commonly in patients with transformed CML and in patients with GIST. ² Headache was the most common in GIST patients. ³ On a patient-year basis, cardiac events including congestive heart failure were more commonly observed in patients with transformed CML than in patients with chronic CML. ⁴ Flushing was most common in GIST patients and bleeding (hematoma, haemorrhage) was most common in patients with GIST and with transformed CML (CML-AP and CML- BC). ⁵ Pleural effusion was reported more commonly in patients with GIST and in patients with transformed CML (CML-AP and CML-BC) than in patients with chronic CML. ^{6/7} Abdominal pain and gastrointestinal haemorrhage were most commonly observed in GIST patients. ⁸ Musculoskeletal pain and related events were more commonly observed in patients with CML than in GIST patients. ⁹ Some fatal cases of hepatic failure and hepatic necrosis have been reported.	

Adverse reactions from post-marketing reports

Infections and infestation	
Not known:	Hepatitis B reactivation
Nervous system disorders	
Uncommon:	Cerebral oedema
Eye disorders	
Rare:	Vitreous haemorrhage
Cardiac disorders	
Rare:	Pericarditis, cardiac tamponade
Vascular disorders	
Uncommon:	Thrombosis/embolism
Rare:	Anaphylactic shock
Respiratory, thoracic and mediastinal disorders	
Uncommon:	Acute respiratory failure ¹ , interstitial lung disease
Gastrointestinal disorders	
Uncommon:	Ileus/intestinal obstruction, tumour haemorrhage/tumour necrosis, gastrointestinal perforation ²
Rare:	Diverticulitis, gastric antral vascular ectasia (GAVE)
Skin and subcutaneous tissue disorders	
Uncommon:	Palmar-plantar erythrodysaesthesia syndrome
Rare:	Lichenoid keratosis, lichen planus
Very Rare:	Toxic epidermal necrolysis
Not known:	Drug rash with eosinophilia and systemic symptoms (DRESS), pseudoporphyria
Musculoskeletal and connective tissue disorders	
Very common:	Musculoskeletal pain upon treatment discontinuation (including myalgia, pain in extremity, arthralgia, bone pain, spinal pain)
Rare:	Avascular necrosis/hip osteonecrosis, rhabdomyolysis/myopathy
Not known:	Growth retardation in children
Reproductive disorders	
Very rare:	Haemorrhagic corpus luteum / haemorrhagic ovarian cyst
Neoplasm benign, malignant and unspecified (including cysts and polyps)	
Rare:	Tumour lysis syndrome
Blood and lymphatic system disorders	
Rare:	Thrombotic microangiopathy
¹ Fatal cases have been reported in patients with advanced disease, severe infections, severe neutropenia and other serious concomitant conditions ² Some fatal cases of gastrointestinal perforation have been reported	

SYMPTOMS AND TREATMENT OF OVERDOSE

Experience with doses higher than the recommended therapeutic dose is limited. Isolated cases of imatinib overdose have been reported spontaneously and in the literature. Generally, the reported outcome in these cases was improvement or recovery. In the event of overdosage the patient should be observed and appropriate symptomatic treatment should be given.

Events that have been reported at different dose ranges are as follows:

Adult overdose:

1200 to 1600 mg (duration varying between 1 to 10 days): Nausea, vomiting, diarrhoea, rash, erythema, oedema, swelling, fatigue, muscle spasms, thrombocytopenia, pancytopenia, abdominal pain, headache, decreased appetite.

1800 to 3200 mg (as high as 3200 mg daily for 6 days): Weakness, myalgia, increased creatine phosphokinase, increased bilirubin, gastrointestinal pain.

6400 mg (single dose): One case reported in the literature of one patient who experienced nausea, vomiting, abdominal pain, pyrexia, facial swelling, decreased neutrophil count, increased transaminases.

8 to 10 g (single dose): Vomiting and gastrointestinal pain have been reported.

Paediatric overdose:

One 3-year-old male exposed to a single dose of 400 mg experienced vomiting, diarrhoea and anorexia and another 3-year-old male exposed to a single dose of 980 mg experienced decreased white blood cell count and diarrhoea.

EFFECTS ON ABILITY TO DRIVE AND USE MACHINES

Reports of motor vehicle accidents have been received in patients receiving imatinib tablets. While most of these reports are not suspected to be caused by imatinib, patients should be advised that they may experience undesirable effects such as dizziness, blurred vision or somnolence during treatment with IMATERO. Therefore, caution should be recommended when driving a car or operating machinery.

INSTRUCTIONS FOR USE

The prescribed dose should be administered orally with a meal and a large glass of water to minimize the risk of gastrointestinal disturbances.

PHARMACEUTICAL PARTICULARS

List of excipients:

Tablet core: Magnesium stearate.

Tablet film coat: Opadry White 03F58763 (Composition: Hypromellose (2910), Titanium dioxide (E171), Macrogol (E1521), Tale (E553b)).

Storage condition

Store below 30°C. Store in the original package. Protect from moisture.

Packaging available:

IMATERO 100: 6 x 10s

IMATERO 400: 3 x 10s

Manufactured by:

Hetero Labs Limited,

Unit – VI, Sy.No.410-411,

TSIIC Formulation SEZ,

Polepally Village, Jadcherla (Mandal),

Mahaboob Nagar (District), Pin-509301,

Telangana, INDIA.

Product Registration Holder:

Camber Laboratories Sdn. Bhd.,

Unit E-13A-02, Menara SUEZCAP 2, No.2,

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